

## 24 Medical physics

### A · 2026 examination · Concept guide and original exam-style practice

#### Understand the concept

Ultrasound echoes locate boundaries because acoustic impedance mismatch reflects part of a pulse; depth equals speed times echo time divided by two. X-ray attenuation depends on material and thickness. CT reconstructs cross-sections from many projections; PET detects pairs of annihilation photons to map tracer distribution. Doses require benefit-risk assessment.

#### Worked example: follow the reasoning

An ultrasound echo returns after  $80\ \mu\text{s}$  in tissue at  $1500\ \text{m s}^{-1}$ : depth =  $1500 \times 80 \times 10^{-6}/2 = 0.060\ \text{m}$ .

#### Quick check · multiple choice

A Why is ultrasound echo time halved to calculate depth?

[1]

- A The probe halves frequency
- B The echo is absorbed twice
- C Speed halves in tissue
- D The sound makes a return trip

Write your choice: \_\_\_\_\_

#### Structured practice

Answer in the space provided. Show working and units where relevant.

1 An echo returns  $120\ \mu\text{s}$  after emission; sound speed in tissue is  $1540\ \text{m s}^{-1}$ . Find depth and explain factor 2.

[3]

Final answer: \_\_\_\_\_

## Answers and explanation

Quick check: D. The sound makes a return trip

### Question 1 [3 marks]

$d = vt/2 = 1540 \times 120 \times 10^{-6}/2 = 0.0924 \text{ m}$  [2]; measured time includes outward and return travel [1].

#### Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Physics 9702, 2025–2027. <https://www.cambridgeinternational.org/Images/664565-2025-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.